

Dra. Citlalli Almaguer Gómez

Professor–Researcher · Photonics, Visual Optics & Science Outreach

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ResearchGate

National Researcher System · Candidate (SECIHTI) 2019–2023 · PRODEP Desirable Profile



PROFILE

I hold a PhD in Photonics and Laser Technologies from the University of Vigo (Spain) and a BSc in Physics from the University of Guadalajara. I am a professor–researcher in the Department of Electro-Photonic Engineering at the University Center for Exact Sciences and Engineering (CUCEI), University of Guadalajara. My research focuses on the design and validation of advanced optical elements —particularly coded phase plates and masks (cubic, trefoil and Jacobi–Fourier)— applied to the correction of complex refractive errors such as presbyopia, astigmatism, keratoconus and post-refractive-surgery conditions.

I lead and collaborate in nationally and internationally funded projects in visual optics, photonic engineering and health-applied sciences. In parallel, I maintain a research line in atmospheric sciences, taking part in studies of climate change, stratospheric dynamics and atmospheric optics, with an emphasis on the interplay between physical processes and their technological applications.

On the educational and social side, I am the founder and coordinator of the Tlakakini Science & Technology Club, an outreach program for children aged 6 to 12 held every month at the University of Guadalajara, which has built a community of more than 160 registered participants. I believe the best way to nurture a scientific vocation is to share science from childhood.

RESEARCH INTERESTS

Visual optics & correction of refractive errors · Coded phase masks & plates (cubic, trefoil, Jacobi–Fourier) · Wavefront coding · Aberrometry & Shack–Hartmann sensors · Photonic engineering & applied optics · Atmospheric sciences & climate change · Stratospheric dynamics & sudden stratospheric warmings · Atmospheric optics · Science outreach & education · Generative AI in science teaching

EDUCATION

- 2013 – 2017** **PhD in Photonics and Laser Technologies**
Universidad de Vigo · Spain
Research on highly aberrated phase elements (cubic phase masks) for presbyopia and astigmatism correction.
- 2011 – 2012** **MSc in Photonics and Laser Technology**
Universidad de Vigo · Spain
Training in optics, photonics and laser technology (Ourense, Spain).
- 2004 – 2010** **BSc in Physics**
Universidad de Guadalajara · Mexico
Foundation in physics and optics that shaped her path toward photonics.

EXPERIENCE

- 2019 – present** **Associate Research Professor B**
Universidad de Guadalajara · CUCEI, Depto. de Ingeniería Electro-Fotónica · Guadalajara, Jalisco · Mexico
Professor with PRODEP Desirable Profile recognition and member of the National Researcher System (2019–2023). Founder and coordinator of the CUCEI Tlakakini Science & Technology Club. Teaches courses in the Photonic Engineering program.
Courses: Nonlinear Optics ×11 · Optical Characterization Laboratory ×9 · Electromagnetic Theory ×6 · Selected Topics in Communications ×6 · Introduction to Engineering ×6 · Lasers ×3 · Measurement Systems

PUBLICATIONS

- 2025** **Astronomical Observation in Mexico Transitioning from the Visible to the Invisible: A Brief Historical Overview**
Almaguer-Gómez C., Almaguer-Medina J. F., Chávez-Aquino G., Chávez-Pérez V. M., et al. · *Óptica Pura y Aplicada* 58(1) · DOI: [10.7149/OPA.58.1.51200](https://doi.org/10.7149/OPA.58.1.51200)

- 2025 Influence of Satellite and Presatellite Periods on the Validation of Major Sudden Stratospheric Warmings in Historical CMIP5 Simulations**
Chávez-Pérez V. M., Añel J. A., Almaguer-Gómez C., de la Torre Ramos L. · *Atmosphere* 16(5):628 · DOI: [10.3390/atmos16050628](https://doi.org/10.3390/atmos16050628)
- 2025 Temporal Variability of Major Stratospheric Sudden Warmings in CMIP5 Climate Change Scenarios**
Chávez-Pérez V. M., Añel J. A., Almaguer-Gómez C., de la Torre Ramos L. · *Climate* 13(10):207 · DOI: [10.3390/cli13100207](https://doi.org/10.3390/cli13100207)
- 2023 Characterization of a Shack–Hartmann Type Wavefront Sensor for Its Use in an Experimental Aberrometer**
Medina-Coca C. M., Sánchez-Valencia B. D., González-Díaz E. A., Almaguer-Gómez C., Magaña-Chávez J. L., Balderas-Mata S. E. · *Proceedings of SPIE* · DOI: [10.1117/12.2676429](https://doi.org/10.1117/12.2676429)
- 2023 Caracterización de un divisor de haz para su utilización en un aberrómetro experimental**
Medina-Coca C. M., Sánchez-Valencia B. D., González-Díaz E. A., Balderas-Mata S. E., Almaguer-Gómez C., Medina-Márquez J. · *SECIHTI / Centro de Investigaciones en Óptica* · ISBN: 978-607-8821-32-7
- 2018 Pupil Size Stability of the Cubic Phase Mask Solution for Presbyopia**
Almaguer-Gómez C., Acosta E., Arines J. · *Journal of Biomedical Optics* 23(1):015002 · DOI: [10.1117/1.JBO.23.1.015002](https://doi.org/10.1117/1.JBO.23.1.015002)
- 2017 Use of a Cubic Phase Plate as Solution for Moderate Astigmatism: Preliminary Study**
Almaguer-Gómez C., Arines J. · *Óptica Pura y Aplicada* 50(2) · DOI: [10.7149/OPA.50.2.49014](https://doi.org/10.7149/OPA.50.2.49014)
- 2017 Potential Use of Cubic Phase Masks for Extending the Range of Clear Vision in Presbyopes: Initial Calculation and Simulation Studies**
Arines J., Almaguer-Gómez C., Acosta E. · *Ophthalmic and Physiological Optics* 37(2) · DOI: [10.1111/opo.12348](https://doi.org/10.1111/opo.12348)
- 2017 Wavefront Coding for Visual Optics**
Acosta E., Arines J., Almaguer-Gómez C. · *Proceedings of SPIE* · DOI: [10.1117/12.2272065](https://doi.org/10.1117/12.2272065)
- 2017 Highly Aberrated Phase Elements for Presbyopia and Astigmatism Correction**
Almaguer-Gómez C., Arines J., Acosta E. · *22nd Microoptics Conference (MOC'17), IEEE* · DOI: [10.23919/MOC.2017.8244539](https://doi.org/10.23919/MOC.2017.8244539)
- 2016 Is Wavefront Coding an Alternative to Adaptive Optics for Retinal Imaging?**
Acosta E., Arines J., Almaguer-Gómez C., Bosch S., Vallmitjana S. · *15th Workshop on Information Optics (WIO), IEEE Xplore* · DOI: [10.1109/WIO.2016.7745563](https://doi.org/10.1109/WIO.2016.7745563)
- 2015 Spreading Optics in the Primary School**
Gargallo A., Gómez-Varela A. I., González-Núñez H., Delgado T., Almaguer-Gómez C., et al. · *Journal of Physics: Conference Series* 605:012040 · DOI: [10.1088/1742-6596/605/1/012040](https://doi.org/10.1088/1742-6596/605/1/012040)
- 2014 Optics Activity for Hospitalized Children**
Gargallo A., Gómez-Varela A. I., González-Núñez H., Delgado T., Almaguer-Gómez C., et al. · *Proceedings of SPIE (ETOP)* · DOI: [10.1117/12.2062820](https://doi.org/10.1117/12.2062820)
- 2014 The USC-OSA Student Chapter: Goals and Benefits for the Optics Community**
Gómez-Varela A. I., Gargallo A., González-Núñez H., Delgado T., Almaguer-Gómez C., et al. · *Proceedings of SPIE (ETOP)* · DOI: [10.1117/12.2070767](https://doi.org/10.1117/12.2070767)

CONFERENCES & TALKS

- 2020 LXIII Congreso Nacional de Física (SMF)**
1–6 nov 2020 · Online
 - Impacto en la frecuencia de ocurrencia de los calentamientos súbitos estratosféricos principales en distintos escenarios de cambio climático usando CMIP5
 - Impacto en las significancias de las características básicas de los SSW al comparar modelos del CMIP5 con datos de reanálisis en periodos pre-satelital y pos-satelital
- 2018 Reunión Anual de la Unión Geofísica Mexicana (RAUGM)**
28 oct–2 nov 2018 · Puerto Vallarta, Mexico
 - Evaluación de la capacidad de reproducción de los calentamientos súbitos estratosféricos principales en el conjunto de datos históricos de CMIP5
 - Calentamientos súbitos estratosféricos principales bajo escenarios de cambio climático con CMIP5
- 2017 22nd Microoptics Conference (MOC'17)**
19–22 nov 2017 · Japan
 - Highly aberrated phase elements for presbyopia and astigmatism correction

- 2017** **Third International Conference on Applications of Optics and Photonics (AOP)**
8–12 may 2017 · Portugal
• Wavefront coding for visual optics
- 2016** **LIX Congreso Nacional de Física (SMF)**
2–7 oct 2016 · León, Mexico
• Teleconexión estratosférica y los calentamientos súbitos estratosféricos
• Patrones característicos en las variables estratosféricas de la Oscilación Cuasi-Bienal
• Anomalías estratosféricas antes y después de los calentamientos súbitos estratosféricos para las distintas fases de la Oscilación Cuasi-Bienal
- 2016** **15th Workshop on Information Optics (WIO)**
11–15 jul 2016 · Spain
• Is wavefront coding an alternative to adaptive optics for retinal imaging?
- 2015** **IX Reunión Nacional de Óptica · Sociedad Española de Óptica**
1–4 sep 2015 · Spain
• Pancorrección de astigmatismo en combinación con miopía, hipermetropía y presbicia mediante el uso de una lámina de fase cúbica
- 2014** **23rd Congress of the International Commission for Optics (ICO-23)**
26–29 ago 2014 · Spain
• Cubic phase-mask for presbyopia correction
• Retinal imaging with wavefront coding
- 2014** **7th European Meeting on Visual and Physiological Optics (EMVPO)**
25–27 ago 2014 · Poland
• Recent advances in retinal imaging with wavefront coding systems
- 2013** **12th International Conference on Education and Training in Optics and Photonics (ETOP)**
23–26 jul 2013 · Portugal
• The USC-OSA Student Chapter: goals and benefits for the optics community
- 2009** **LII Congreso Nacional de Física (SMF)**
26–30 oct 2009 · Acapulco, Mexico
• Espectros de frecuencias de datos meteorológicos de Isla María Madre

ACTIVE RESEARCH PROJECTS

- 2026 – 2028** **Design and validation of optical elements to enhance visual acuity through wavefront coding — Technical lead / PI**
CUCEI · Universidad de Guadalajara · Funding: Project with institutional funding (UDG)
Technical lead. Development and evaluation of a new generation of optical elements that extend depth of focus and compensate for refractive irregularities of the human eye using coded phase masks (cubic, trefoil and Jacobi–Fourier). The work spans numerical simulation in MatLab, experimental validation with a spatial light modulator (SLM) and CCD camera, and the fabrication of customized contact lenses.
- 2024 – 2025** **Study of sudden stratospheric warmings using CMIP5 — Collaborator**
Universidad de Guadalajara · Universidade de Vigo (colaboración internacional) · Funding: International collaboration
Collaborator in an international project on the variability of sudden stratospheric warmings in CMIP5 simulations, contributing to methodological design, variable definition and integration of historical data.
- 2025 – actual** **WWLLN lightning detection station (UDG – University of Washington) — Institutional technical contact**
CUTLAQUE · Universidad de Guadalajara · Funding: Interinstitutional collaboration agreement
Institutional technical contact for the Mexican station of the World Wide Lightning Location Network (WWLLN), a global scientific network led by Dr. Robert H. Holzworth (University of Washington). Installed in 2025, the station improves lightning-detection efficiency over thousands of kilometers and generates open-access data for atmospheric research.

TECH DEVELOPMENTS

Tlakakini web system for science-fair management

HTML5 · CSS3 · JavaScript · Google Apps Script · Google Sheets

Web application for the end-to-end management of the UDG Tlakakini Club science fairs. It coordinates administrators, workshop leaders and participants through modules for registration, workshop proposal and approval, enrollment with schedule validation, real-time attendance tracking and a six-category gamified system. Serverless architecture. Registered with INDAUTOR and licensed to CU Tlaquepaque (UDG).

Weather forecasting system · CU Tlaquepaque

HTML5 · CSS3 · JavaScript · GFS (NOAA) · NASA GIBS · Leaflet

Real-time weather forecasting web application for Mexico. It visualizes GFS 0.25° numerical-model data (NOAA/NOMADS) and satellite temperature (NASA GIBS/MODIS) on an interactive map with layers for temperature, precipitation, humidity, wind, UV index, air quality and atmospheric electrical activity (WWLLN). It generates an automated daily weather bulletin and visual alerts. Registered with INDAUTOR.

COMMUNITY BUILDING & SERVICE

Thesis supervision

- 2025** **Demostración del efecto Rayleigh y Mie con pinzas ópticas**
Aldo Josué Chávez Gómez · Bachelor's · Advisor · Universidad de Guadalajara
- 2025** **Diseño de un microscopio funcional con fuente de iluminación LED para el análisis y procesamiento de imágenes 3D**
Luz Athziri Molin Gómez · Bachelor's · Reviewer · Universidad de Guadalajara
- 2025** **Caracterización del desempeño de un prototipo de sensor de onda tipo Shack–Hartmann e incremento en la versatilidad de su software**
Brandon Daniel Sánchez Valencia · Bachelor's · Reviewer · Universidad de Guadalajara
- 2025** **Caracterización del desempeño de un prototipo de sensor de onda tipo Shack–Hartmann e incremento en la versatilidad de su software**
Cristian Manuel Medina Coca · Bachelor's · Reviewer · Universidad de Guadalajara

Review & assessment

- 2025** **Comité de Titulación · Ing. Fotónica, CUCEI–UDG**
Reviewer of modular graduation projects (2025A and 2025B terms)
- 2025** **Comité Técnico · Ing. Informática, CUCEI–UDG**
Reviewer of modular graduation projects

Project evaluation

- 2020** **SECIHTI (CONACYT)**
External evaluator — 2020 Postdoctoral Stays in Mexico Call (2 applications)

SCIENCE OUTREACH

- 2024 –** **Tlakakini Science & Technology Club**
Founder and coordinator. Monthly science sessions for children aged 6–12 at CUCEI; a community of 160+ registered participants.
- 2024 –** **Tlakakini Science Fridays**
Coordination and delivery of hands-on workshops for children: atmospheric pressure, non-Newtonian fluids, colored shadows, photosynthesis and more.
- 2025** **UDGTV Channel 44 interview about the Tlakakini Club**
Television feature on the outreach model of the Tlakakini Science Club.
- 2025** **“Learning science is an act of rebellion”**
Talk for the International Day of Women and Girls in Science (COECYTJAL): pioneers who broke gender barriers.
- 2026** **“Women who inspire: from curiosity to science”**
Panel for the International Day of Women and Girls in Science at the University of Guadalajara.
- 2021** **Astronomy and optics workshops**
“Galaxies”, “Phases of the Moon” and “Behind the title” workshops at IAM-UdeG and World Space Week.

GRANTS & HONORS

- 2019–2023** **National Researcher System · Candidate**
SECIHTI · Mexico
- 2020** **Desirable Profile Recognition & Grant (PRODEP)**
SEP (PRODEP) · Mexico
- 2015** **Doctoral Studies Scholarship**
Universidad de Guadalajara · Mexico
- 2011** **«Ourense Exterior» Master's Scholarship**
Diputación de Ourense · Universidad de Vigo · Spain

CONTINUING EDUCATION

- 2025** **Diploma** — Estrategias tecnopedagógicas para el desarrollo de habilidades del siglo XXI (DETHAS) · Universidad de Guadalajara
- 2025** **Course** — Generative AI Leader · Google Cloud
- 2025** **Course** — IA Generativa en el aula · UNAM
- 2025** **Course** — Inteligencia Artificial Generativa para Docentes con ChatGPT · Universidad Anáhuac
- 2025** **Course** — University Teaching · The University of Hong Kong
- 2025** **Course** — Lengua de Señas Mexicana · Universidad de Guadalajara
- 2024** **Course** — Introducción a las simulaciones PhET para la educación STEM · University of Colorado Boulder
- 2024** **Course** — Tesoros de la Física y sus descubridores I · Universidad de los Andes
- 2022** **Diploma** — Tutorías · Universidad de Guadalajara
- 2021** **Diploma** — Diseño de unidades de aprendizaje para la modalidad en línea · Universidad de Guadalajara
- 2018** **Diploma** — Competencias docentes: inducción al bachillerato general por competencias · Universidad de Guadalajara
- 2014** **Course** — International Summer School: Frontiers on Photonics and Laser Technologies · Universidad de Santiago de Compostela

LANGUAGES

Spanish — Native · English — Intermediate · Galician — Basic

TOOLBOX

MatLab · Python · SLM · Shack–Hartmann · CCD · LaTeX · HTML5 / CSS3 / JS · Google Apps Script · PhET